

# ANX-PR/CL/001-01

## LEARNING GUIDE

### SUBJECT

**103000389 - Data Mining**

### DEGREE PROGRAMME

**10AK - Master Universitario En Software Y Sistemas**

### ACADEMIC YEAR & SEMESTER

**2025/26 - Semester 2**

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### Learning guide

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## 1. Description

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### 1.1. Subject details

|                                       |                                                    |
|---------------------------------------|----------------------------------------------------|
| <b>Name of the subject</b>            | 103000389 - Data Mining                            |
| <b>No of credits</b>                  | 4 ECTS                                             |
| <b>Type</b>                           | Optional/elective                                  |
| <b>Academic year of the programme</b> | First year                                         |
| <b>Semester of tuition</b>            | Semester 2                                         |
| <b>Tuition period</b>                 | February-June                                      |
| <b>Tuition languages</b>              | English                                            |
| <b>Degree programme</b>               | 10AK - Master Universitario en Software y Sistemas |
| <b>Centre</b>                         | 10 - E.T.S. De Ingenieros Informáticos             |
| <b>Academic year</b>                  | 2025-26                                            |

## 2. Faculty

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### 2.1. Faculty members with subject teaching role

| <b>Name and surname</b>                           | <b>Office/Room</b> | <b>Email</b>                | <b>Tutoring hours *</b>                         |
|---------------------------------------------------|--------------------|-----------------------------|-------------------------------------------------|
| Fco.javier Segovia Perez<br>(Subject coordinator) | 2305               | javier.segovia@upm.es       | M - 10:00 - 11:00<br>Hablar con el<br>profesor  |
| Ernestina Menasalvas Ruiz                         | 4303               | ernestina.menasalvas@upm.es | M - 10:00 - 11:00<br>hablar con la<br>profesora |

\* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

### 3. Prior knowledge recommended to take the subject

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#### 3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

#### 3.2. Other recommended learning outcomes

- Artificial Intelligence
- Statistics

### 4. Skills and learning outcomes \*

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#### 4.1. Skills to be learned

CEM7 - Evaluar y aplicar las diversas teorías matemáticas y estadísticas, y los procesos, métodos y técnicas disponibles para la extracción y descubrimiento de conocimiento a partir de grandes volúmenes de datos

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio.

CG3 - Que los estudiantes sepan comunicar sus conclusiones y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades.

## 4.2. Learning outcomes

RA135 - Obtain data mining results from descriptive, diagnostic, predictive, and prescriptive analysis by applying a broad and varied repertoire of classification, regression, association, segmentation, and clustering algorithms.

RA134 - Turn a business question into a data question, and use what is learned from applying a data mining process to make smart decisions that help the company or institution succeed.

\* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

## 5. Brief description of the subject and syllabus

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### 5.1. Brief description of the subject

At the end of this course we will make you able to:

- Turn a business question into a data question, and use what is learned from applying a data mining process to make smart decisions that help the company or institution succeed.
- Obtain data mining results from descriptive, diagnostic, predictive, and prescriptive analysis by applying a broad and varied repertoire of classification, regression, association, segmentation, and clustering algorithms.

We will use a professional data mining tool called IBM SPSS Modeler, which is designed to be user-friendly and does not require any prior programming experience.

You will learn how to use this tool by following an IBM tutorial, along with interactive class exercises and assignments.

This course is very hands-on and practical?you will mainly learn by doing. Through real examples and guided practice, you'll gain the skills needed to analyze data effectively.

By the end of the course, you will be able to solve a real-world problem of your choice using data mining techniques. Throughout the course, we will go through all the key phases of data analysis, step by step, while

practicing with real-life examples.

## 5.2. Syllabus

1. INTRODUCTION TO DATA ANALYSIS AND THE IBM SPSS MODELER
2. BUSINESS VALUE PROPOSITION & DATA MINING
3. DESCRIPTIVE ANALYSIS USING BASIC STATISTICS
4. DESCRIPTIVE ANALYSIS: DATA VISUALIZATION
5. DESCRIPTIVE ANALYSIS: RFM
6. DESCRIPTIVE ANALYSIS: CLUSTERING
7. DIAGNOSTIC ANALYSIS: CORRELATION, ANOVA AND CHI-SQUARED TESTS
8. DIAGNOSTIC ANALYSIS: ASSOCIATION RULES
9. PREDICTIVE ANALYSIS: LINEAR REGRESSION
10. PREDICTIVE ANALYSIS: LOGISTIC REGRESSION
11. PREDICTIVE ANALYSIS: DECISION TREES
12. PREDICTIVE ANALYSIS: NEAREST NEIGHBOR
13. PREDICTIVE ANALYSIS: NEURAL NETWORKS
14. PREDICTIVE ANALYSIS: PROBABILISTIC MODELS
15. PREDICTIVE ANALYSIS: ENSEMBLE METHODS
16. PREDICTIVE ANALYSIS: DEALING WITH TIME
17. PRESCRIPTIVE ANALYSIS: TARGETING CUSTOMERS
18. PRESCRIPTIVE ANALYSIS: RECOMMENDATION SYSTEMS
19. FINAL PROJECT

## 6. Schedule

### 6.1. Subject schedule\*

| Week | Type 1 activities                                                                                                                                                                                                                                                                                                                                   | Type 2 activities | Distant / On-line | Assessment activities                                                                                                                                                                                                              |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Introduction to Data Mining</b><br>Duration: 00:30<br>Lecture<br><br><b>Exercise about Value Proposition Canvas</b><br>Duration: 00:30<br>Problem-solving class<br><br><b>Problem about Data Analysis on a Business Case</b><br>Duration: 00:30<br>Problem-solving class<br><br><b>Tool practice</b><br>Duration: 02:30<br>Problem-solving class |                   |                   | <b>Tool practice</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 02:30<br><br><b>Theoretical Problem</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 01:00 |
| 2    | <b>Problem about DATA VISUALIZATION using the tool</b><br>Duration: 02:00<br>Problem-solving class<br><br><b>Problem about STATISTICAL DESCRIPTION using the tool</b><br>Duration: 02:00<br>Problem-solving class                                                                                                                                   |                   |                   | <b>Practical Problem using the tool</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 04:00                                                                                                      |
| 3    | <b>Problem about RFM using the tool</b><br>Duration: 02:00<br>Problem-solving class<br><br><b>Problem about CLUSTERING using the tool</b><br>Duration: 02:00<br>Problem-solving class                                                                                                                                                               |                   |                   | <b>Practical Problem using the tool</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 04:00                                                                                                      |
| 4    | <b>Problem about CORRELATION, ANOVA AND CHI-SQUARED TESTS using the tool</b><br>Duration: 02:00<br>Problem-solving class<br><br><b>Problem about ASSOCIATION RULES using the tool</b><br>Duration: 02:00<br>Problem-solving class                                                                                                                   |                   |                   | <b>Practical Problem using the tool</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 04:00                                                                                                      |

|    |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5  | <b>Problem about LINEAR REGRESSION using the tool</b><br>Duration: 02:00<br>Problem-solving class<br><br><b>Problem about LOGISTIC REGRESSION using the tool</b><br>Duration: 02:00<br>Problem-solving class               |  |  | <b>Practical Problem using the tool</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 04:00                                                                                                                                                                                |
| 6  | <b>Problem about NEAREST NEIGHBOR AND NEURAL NETWORKS USING THE TOOL</b><br>Duration: 02:00<br>Problem-solving class<br><br><b>Problem about DECISION TREES using the tool</b><br>Duration: 02:00<br>Problem-solving class |  |  | <b>Practical Problem using the tool</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 04:00                                                                                                                                                                                |
| 7  | <b>Problem about TIME RELATED CASES using the tool</b><br>Duration: 02:00<br>Problem-solving class<br><br><b>Problem about ENSEMBLE METHODS using the tool</b><br>Duration: 02:00<br>Problem-solving class                 |  |  | <b>Practical Problem using the tool</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 04:00                                                                                                                                                                                |
| 8  | <b>Problem about PRESCRIPTIVE ANALYSIS using the tool</b><br>Duration: 02:00<br>Problem-solving class                                                                                                                      |  |  | <b>Practical Problem using the tool</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 02:00                                                                                                                                                                                |
| 9  |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 10 |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 11 |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 12 |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 13 |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 14 |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 15 |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 16 |                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                              |
| 17 | <b>FINAL PROJECT PRESENTATION</b><br>Duration: 00:20<br>Additional activities                                                                                                                                              |  |  | <b>FINAL PROJECT presentation and all assignments uploaded</b><br>Individual presentation<br>Progressive assessment<br>Presential<br>Duration: 00:20<br><br><b>FINAL PROJECT presentation and all assignments uploaded</b><br>Individual presentation<br>Global examination<br>Presential<br>Duration: 00:20 |

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.



## 7. Activities and assessment criteria

### 7.1. Assessment activities

#### 7.1.1. Assessment

| Week | Description                                             | Modality                | Type         | Duration | Weight | Minimum grade | Evaluated skills   |
|------|---------------------------------------------------------|-------------------------|--------------|----------|--------|---------------|--------------------|
| 1    | Tool practice                                           | Individual presentation | Face-to-face | 02:30    | 1%     | 5 / 10        | CG1                |
| 1    | Theoretical Problem                                     | Individual presentation | Face-to-face | 01:00    | 1%     | 5 / 10        | CG1<br>CG3         |
| 2    | Practical Problem using the tool                        | Individual presentation | Face-to-face | 04:00    | 1%     | 5 / 10        | CEM7<br>CG1<br>CG3 |
| 3    | Practical Problem using the tool                        | Individual presentation | Face-to-face | 04:00    | 1%     | 5 / 10        | CEM7<br>CG1<br>CG3 |
| 4    | Practical Problem using the tool                        | Individual presentation | Face-to-face | 04:00    | 1%     | 5 / 10        | CEM7<br>CG1<br>CG3 |
| 5    | Practical Problem using the tool                        | Individual presentation | Face-to-face | 04:00    | 1%     | 5 / 10        | CEM7<br>CG1<br>CG3 |
| 6    | Practical Problem using the tool                        | Individual presentation | Face-to-face | 04:00    | 1%     | 5 / 10        | CEM7<br>CG1<br>CG3 |
| 7    | Practical Problem using the tool                        | Individual presentation | Face-to-face | 04:00    | 1%     | 5 / 10        | CEM7<br>CG1<br>CG3 |
| 8    | Practical Problem using the tool                        | Individual presentation | Face-to-face | 02:00    | 1%     | 5 / 10        | CEM7<br>CG1<br>CG3 |
| 17   | FINAL PROJECT presentation and all assignments uploaded | Individual presentation | Face-to-face | 00:20    | 91%    | 5 / 10        | CEM7<br>CG1<br>CG3 |

#### 7.1.2. Global examination

| Week | Description | Modality | Type | Duration | Weight | Minimum grade | Evaluated skills |
|------|-------------|----------|------|----------|--------|---------------|------------------|
|------|-------------|----------|------|----------|--------|---------------|------------------|

|    |                                                         |                         |              |       |      |        |                    |
|----|---------------------------------------------------------|-------------------------|--------------|-------|------|--------|--------------------|
| 17 | FINAL PROJECT presentation and all assignments uploaded | Individual presentation | Face-to-face | 00:20 | 100% | 5 / 10 | CEM7<br>CG1<br>CG3 |
|----|---------------------------------------------------------|-------------------------|--------------|-------|------|--------|--------------------|

### 7.1.3. Referred (re-sit) examination

| Description                                             | Modality                | Type         | Duration | Weight | Minimum grade | Evaluated skills   |
|---------------------------------------------------------|-------------------------|--------------|----------|--------|---------------|--------------------|
| FINAL PROJECT presentation and all assignments uploaded | Individual presentation | Face-to-face | 00:20    | 100%   | 5 / 10        | CEM7<br>CG1<br>CG3 |

## 7.2. Assessment criteria

Course Evaluation Overview:

- In every session, we will work on a problem or assignment in class. Your final grade is based on the assignments from each session, and mainly, your final project.
- Important: You must complete, upload, and pass (minimum grade of 5 out of 10) all assignments to pass the course.

Types of Evaluation

#### 1. Continuous (Progressive) Evaluation

You can resubmit each assignment once if needed.

#### 2. Global Evaluation

You can only submit each assignment once. No resubmissions allowed.

#### 3. Extraordinary Evaluation

You may resubmit only the assignments you failed during the continuous or global evaluations.

## 8. Teaching resources

### 8.1. Teaching resources for the subject

| Name                                                                                                                                                         | Type         | Notes                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------|
| Principles of Data Mining (Adaptive Computation and Machine Learning), D Hand, MIT Press, 2001.                                                              | Bibliography |                       |
| Jiawei Han, Micheline Kamber, Data Mining : Concepts and Techniques, 2nd edition, Morgan Kaufmann, ISBN 1558609016, 2006.                                    | Bibliography |                       |
| Data Mining Techniques: Marketing, Sales and Customer Support, Michael J. A. Berry, Gordon Linoff, John Wiley & Sons, 1997.                                  | Bibliography |                       |
| Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Introduction to Data Mining, Pearson Addison Wesley (May, 2005). Hardcover: 769 pages. ISBN: 0321321367       | Bibliography | MOST RECOMMENDED BOOK |
| Ian Witten, Eibe Frank, Mark Hall, Data Mining: Practical Machine Learning Tools and Techniques, 3rd Edition, Morgan Kaufmann, ISBN 978-0-12-374856-0, 2011. | Bibliography |                       |
| Página web de la asignatura en moodle                                                                                                                        | Web resource |                       |
| IBM SPSS MODELER                                                                                                                                             | Others       | THE TOOL WE WILL USE  |
| aula                                                                                                                                                         | Equipment    |                       |

## 9. Other information

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### 9.1. Other information about the subject

Several innovative teaching methodologies are implemented in the course (<https://innovacioneducativa.upm.es/guias-pdi>) with the aim of motivating and reinforcing student learning:

? Learning by Doing: I use this technique, first, to help students learn how to use the IBM SPSS Modeler data mining tool. They follow a tutorial at home to learn the basics, and I continue with my own tutorial during some class sessions. Second, instead of just reading about data mining or doing exercises on paper, in each class session?after I give a theoretical introduction to a concept or technique?students use IBM SPSS Modeler to analyze real data and solve problems related to that concept or algorithm.

? Project based Learning: A large part of the students' final grade comes from completing a data mining project of their choice. Around the middle of the course, students must find a dataset related to a real-world problem that interests them and begin applying a data mining process to solve it, with a final report submitted at the end. I supervise and monitor their progress, answering questions and guiding them when they encounter difficulties. Projects can be of any type. The only requirement I impose is that students must first inform me about the dataset they want to use and the type of problem they aim to solve with it. I then check whether the dataset contains enough information and data to make the project feasible. If I determine that the dataset lacks the necessary information to successfully apply data mining techniques and algorithms, I ask them to choose a different problem and/or dataset.